

Gaston Elian Bujia

Research Data Scientist | Machine Learning | Python | Computational Modeling

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Industry Profile

Data Scientist with a strong background in mathematics, machine learning, and applied research. I combine hands-on experience in data-driven analysis with doctoral training in computational modeling, Bayesian methods, eye-tracking data, and deep learning. I am especially interested in roles where rigorous quantitative thinking, Python-based analysis, and clear communication can turn complex data into useful decisions or research products.

Core Skills

- **Python and data analysis:** Python, NumPy, Pandas, SciPy, Jupyter Notebooks, data cleaning, exploratory analysis, and reproducible analysis workflows.
 - **Machine Learning:** scikit-learn, PyTorch, TensorFlow, deep learning, model evaluation, and applied predictive modeling.
 - **Statistical and computational modeling:** Bayesian modeling, computational neuroscience, eye-tracking data, behavioral data, and simulation-based analysis.
 - **Visualization and communication:** Matplotlib, Seaborn, Plotly, technical writing, teaching, and communication of quantitative results to technical audiences.
 - **Tools:** Git, GitHub, LaTeX, open research software, and collaborative technical workflows.
 - **Languages:** Spanish native; English professional working proficiency.
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Relevant Experience

Research Data Scientist | **Everything ALS** | *March 2024 - Present*

- Data-driven analysis in the context of amyotrophic lateral sclerosis research.
- Applied quantitative modeling and exploratory analysis to complex biomedical research data.

Graduate Researcher | **Applied Artificial Intelligence Lab (UBA-CONICET)** | *2018 - Present*

- Developed and evaluated computational models for eye-movement prediction and visual search in natural scenes.
- Worked with behavioral and eye-tracking data using machine learning, Bayesian modeling, and deep learning methods.
- Designed reproducible analyses for complex experimental data in an applied artificial intelligence research environment.

Open Research Software and Teaching Resources | **GitHub / NeuroLIAA** | *2022 - Present*

- Contributor to **ViSioNS**, a benchmark used to evaluate computational models of human visual search in natural scenes.

- Contributor to **sIBS**, a Bayesian visual-search model for naturalistic stimuli.
 - Published open educational material on data processing and visualization with Python.
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Technical Teaching and Mentoring

Senior Teaching Assistant / Teaching Assistant Department of Computer Science, School of Exact and Natural Sciences, University of Buenos Aires | 2021 - Present

- University teaching in **Machine Learning**, **Computational Statistics**, and computer systems for undergraduate and graduate audiences.
- Experience translating quantitative and computational concepts into clear, structured technical material.

Graduate Teaching Assistant Master's Program in Data Mining and Knowledge Discovery, School of Exact and Natural Sciences, University of Buenos Aires | 2021 - 2024

- Graduate-level teaching in **Machine Learning** for a data mining and knowledge discovery master's program.

Thesis Co-supervisor and Undergraduate Research Mentor University of Buenos Aires | 2021 - Present

- Mentored undergraduate research and thesis projects involving computational modeling, optimization, and behavioral data.
 - Guided junior collaborators through problem formulation, implementation choices, and technical communication.
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Education

- **Ph.D. in Computer Science** | UBA (2018 - Present)
 - Doctoral thesis titled *Computational aspects of human vision: prediction of eye movements during visual search*. Expected defense between July and August 2026. Advisor: Dr. Juan Kamienkowski. Co-advisor: Dr. Guillermo Solovey.
 - **B.Sc. in Mathematics** | UBA (2018)
 - Applied mathematics focus. Thesis: *Recommendation systems: Factorization Machines and the cold-start problem*. Advisor: Dr. Leandro Lombardi.
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Selected Research Output

1. Ruarte, G., **Bujia, G.**, Care, D., Ison, M. J., & Kamienkowski, J. E. (2025). *Integrating Bayesian and neural networks models for eye movement prediction in hybrid search*. In **Scientific Reports**, 15(1), 16482.
2. Dziemian, S., **Bujia, G.**, Prasse, P., Barańczuk-Turska, Z., Jäger, L. A., Kamienkowski, J. E., & Langer, N. (2025). *Saliency models reveal reduced top-down attention in Attention-Deficit/Hyperactivity Disorder: A naturalistic eye-tracking study*. In **JAACAP Open**, 3(2), 192–204.
3. **Bujia, G.**, Ruarte, G., Sclar, M., Solovey, G., & Kamienkowski, J. E. (2025). *Uncertainty during visual search: Insights from a computational model and behavioral experiment in natural stimuli*. In **bioRxiv**.